

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1508

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I, **G. Umesh Chandra Yadav/192525424** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Umesh*

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

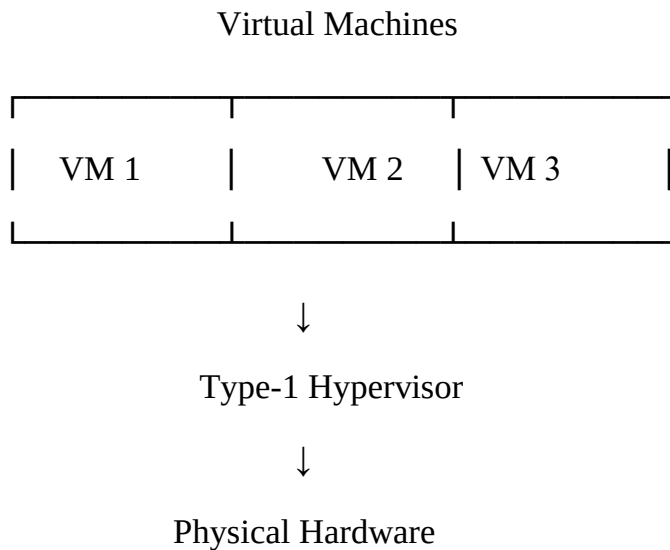
Problem 1: Compare Type-1 and Type-2 Hypervisors for a Mission-Critical Cloud Datacenter

Problem Statement

A cloud datacenter is required to host mission-critical applications where high performance, strong security, availability, scalability, and centralized management are essential. The organization must decide between a Type-1 hypervisor and a Type-2 hypervisor.

1. Type-1 Hypervisor

A Type-1 hypervisor, also called a bare-metal hypervisor, runs directly on the physical server hardware.



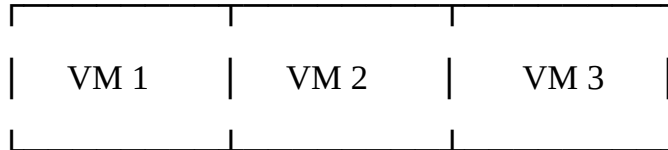
Examples include **VMware ESXi**, Microsoft Hyper-V, and Xen.

Since there is no general-purpose host operating system between the hypervisor and hardware, resources such as CPU, memory, storage and networking can be managed more efficiently.

2. Type-2 Hypervisor

A Type-2 hypervisor runs as an application on top of a conventional host operating system.

Virtual Machines



Type-2 Hypervisor



Host Operating System



Physical Hardware

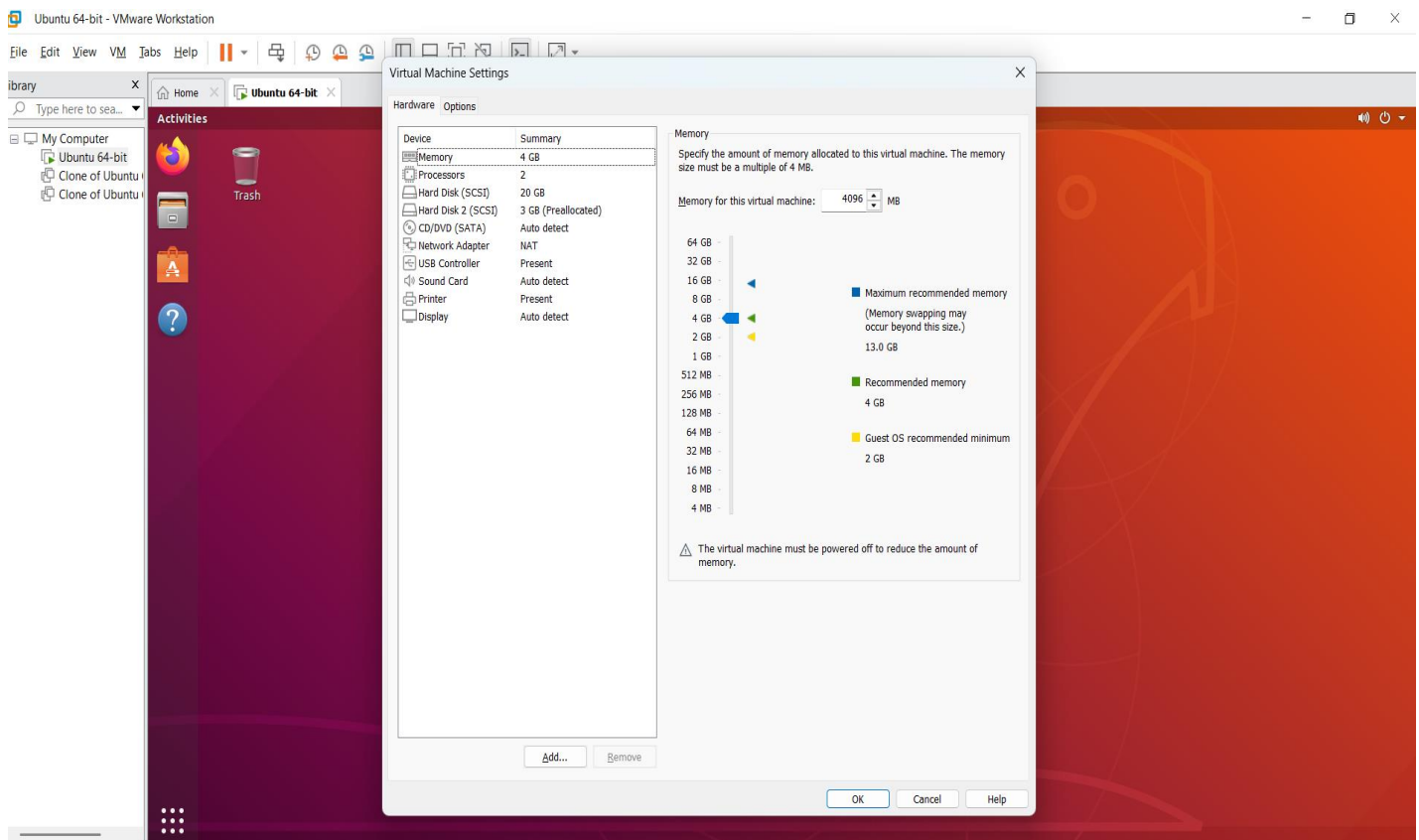
Factor	Type-1 Hypervisor	Type-2 Hypervisor
Architecture	Runs directly on physical hardware	Runs on top of a host operating system
Performance	High performance with low overhead	Lower performance due to host OS overhead
Security	Stronger isolation and smaller attack surface	More dependent on host OS security
Resource Management	Efficient CPU, memory and storage management	Resources are managed through the host OS
Manageability	Centralized and suitable for datacentres	Easier for desktop/testing environments
Scalability	Highly scalable	Limited for large datacentres
Examples	VMware ESXi, Microsoft Hyper-V	VMware Workstation, Oracle VirtualBox

Justification

For a **mission-critical cloud datacenter**, **Type-1 hypervisor is the better choice** because it:

1. Provides higher performance.
2. Reduces virtualization overhead.
3. Provides stronger VM isolation.
4. Supports centralized management.
5. Supports high availability and live migration.
6. Handles large-scale workloads efficiently.

Conclusion: Type-1 hypervisors are more suitable for cloud datacentres hosting mission-critical applications.



PROBLEM 2. Workload Isolation in a Cloud Provider

Problem

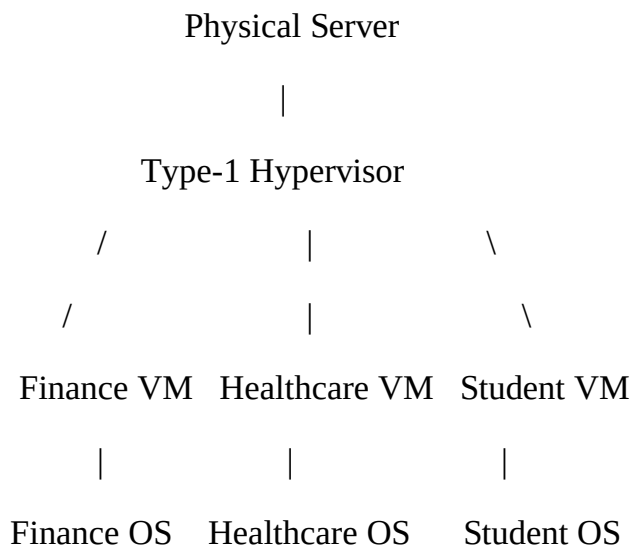
The cloud provider has three workloads:

- Finance
- Healthcare
- Student Portal

All workloads are running on the **same physical hardware**.

How Virtualization Provides Isolation

Virtualization creates separate **Virtual Machines (VMs)** for each workload.



Each VM gets isolated:

- CPU resources
- Memory
- Virtual storage
- Virtual network interface
- Guest operating system

The hypervisor prevents one VM from directly accessing another VM's resources.

Two Major Attack Surfaces

1. Hypervisor Vulnerabilities

If the hypervisor contains a security vulnerability, an attacker could potentially perform **VM escape** and access other VMs or the host.

2. Management Interface / APIs

Cloud management interfaces and APIs are major attack surfaces. Weak passwords, stolen credentials, poor access control, or vulnerable APIs can allow attackers to control VMs and cloud resources.

Security Measures

- Strong VM isolation
- Role-Based Access Control (RBAC)
- Network segmentation
- Secure APIs
- Multi-factor authentication
- Regular hypervisor patching
- Encryption of sensitive data

Conclusion

Virtualization allows finance, healthcare and student workloads to safely share hardware while maintaining logical isolation. However, the **hypervisor and management interfaces** must be strongly secured.

PROBLEM 3. VM Host Utilization Analysis

Given Snapshot

Resource	Utilization
CPU	92%
Memory	88%
Storage I/O Latency	35 ms
Average VM Boot Time	170 seconds

Analysis of Bottlenecks

1. CPU Bottleneck

CPU utilization is **92%**, which is very high.

Possible problems:

- CPU contention between VMs

- Excessive VM workloads
- High context switching
- Reduced application performance

2. Memory Bottleneck

Memory utilization is **88%**, which indicates limited available RAM.

Possible problems:

- Memory swapping
- Memory ballooning
- Increased paging
- Reduced VM performance

3. Storage I/O Bottleneck

Storage I/O latency is **35 Ms**, which is relatively high for many VM workloads.

Possible causes:

- Slow storage devices
- Heavy disk activity
- Too many VMs accessing the same datastore
- Insufficient storage throughput

4. VM Boot Time

Average boot time is **170 seconds**, which is quite high.

The likely causes are:

- High storage I/O latency
- CPU contention
- Memory pressure
- Slow VM image access

Corrective Actions

1. Reduce CPU contention

- Move some VMs to another host.
- Increase CPU capacity if required.

2. Increase available memory

- Add physical RAM.
- Optimize VM memory allocation.
- Reduce unnecessary memory-consuming workloads.

3. Improve storage performance

- Use SSD/NVMe storage.
- Optimize datastore configuration.
- Distribute VMs across storage resources.

4. Reduce VM boot time

- Optimize VM images.
- Remove unnecessary startup services.
- Improve storage performance.

5. Continuous monitoring

- Monitor CPU, RAM and storage regularly.
- Configure alerts when resource usage becomes too high.

Conclusion

The main bottlenecks are **high CPU utilization, high memory utilization and storage I/O latency**. The storage bottleneck is especially likely to contribute to the **170-second VM boot time**.

```
vm-monitor - VMware Workstation
File Edit View VM Tabs Help

cloudadmin@vm-monitor:~$ uptime
10:42:17 up 45 days,  5:21,  3 users,  load average: 2.67, 2.41, 2.10
cloudadmin@vm-monitor:~$ top -b -n 1 | head -n 20
top - 10:42:17 up 45 days,  5:21,  3 users,  load average: 2.67, 2.41, 2.10
Tasks: 236 total,   2 running, 233 sleeping,   1 stopped,   0 zombie
%Cpu(s): 92.0 us,  6.1 sy,  0.3 ni,  1.0 id,  0.2 wa,  0.2 hi,  0.2 si,  0.0 st
MiB Mem : 15974.5 total, 1412.6 free, 1407.3 buff/cache
MiB Swap: 2048.0 total,  245.6 free, 1802.4 used. 14050.3 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 1234 root        20   0 2154320 512432 24576 R   45.6   3.1   12:25.31 qemu-system
 1250 root        20   0 1987652 412300 21244 R   35.2   2.5    9:11.07 kvm
 1300 root        20   0 1123456 210456 15432 S    8.7   1.3    4:25.12 httpd
 1400 cloudadm   20   0  532120  45230 36244 S    2.3   0.3    0:31.55 sshd
 1500 root        20   0  423120  32120 11220 S    1.0   0.2    0:25.10 systemd

cloudadmin@vm-monitor:~$ free -m
              total        used        free      shared  buff/cache   available
Mem:           15974         14085         1412          108          1407        14050
Swap:           2048           1802           246

cloudadmin@vm-monitor:~$ iostat -x 1 1
Device            r/s    w/s  rkB/s  wkB/s  rrqm/s  wrqm/s  %util   await  r_await  w_await  svctm  %util
sda                120.5   98.7 10240.5 20480.3   10.2   15.6   95.4   35.12  10.45   51.32    5.12  95.4

cloudadmin@vm-monitor:~$ █
```

PROBLEM 4. Software Defined Datacenter (SDDC) and Agility

Traditional Datacenter

A traditional datacenter generally depends heavily on physical hardware.

Traditional Datacenter

|

Physical Servers

|

Physical Storage

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Physical Network

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Manual Configuration

This results in:

- Slow provisioning
- Manual configuration
- Hardware dependency
- Difficult scaling
- Higher operational effort

Software Defined Datacenter

An SDDC abstracts datacenter resources using software.

SDDC

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Virtualization Management

/

|

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Compute Storage Network

|

|

|

VMs Virtual SDN

Storage

1. Compute Virtualization

Server virtualization converts physical servers into virtual machines.

Benefits:

- Rapid VM deployment
- Better resource utilization
- Easy scaling
- VM migration

- Automated provisioning

2. Storage Virtualization

Storage virtualization combines physical storage resources into logical storage pools.

Benefits:

- Flexible storage allocation
- Easier expansion
- Better storage utilization
- Automated storage management

3. Network Virtualization

Network virtualization creates logical networks independent of physical network hardware.

Examples include:

- Software Defined Networking (SDN)
- Virtual switches
- Network overlays

Benefits:

- Rapid network configuration
- Easier segmentation
- Automated network provisioning
- Flexible traffic management

How SDDC Improves Agility

1. **Rapid provisioning** – Resources can be created quickly using software.
2. **Automation** – Reduces manual configuration.
3. **Elastic scalability** – Resources can scale according to demand.
4. **Centralized management** – Compute, storage and networking can be managed through software.

5. **Better resource utilization** – Resources can be dynamically allocated.
6. **Faster deployment** – Applications can be deployed faster than in traditional datacenters.

Conclusion:

SDDC improves datacenter agility by converting hardware-based infrastructure into a **software-controlled, automated and programmable environment**.

```
sddcuser@sddc-demo:~$ echo "=== SOFTWARE DEFINED DATACENTER (SDDC) AGILITY ANALYSIS ==="
```

```
=== SOFTWARE DEFINED DATACENTER (SDDC) AGILITY ANALYSIS ===
```

```
1. Compute Virtualization
```

- VMs/Containers can be provisioned in seconds.
- Resource pooling enables dynamic allocation.
- Automation reduces manual effort and errors.
- Improves scalability and workload mobility.

```
2. Storage Virtualization
```

- Storage pooling abstracts physical storage.
- Thin provisioning enables efficient usage.
- Storage can be provisioned on-demand.
- Improves flexibility and disaster recovery.

```
3. Network Virtualization
```

- Virtual networks (VLAN/VXLAN) created instantly.
- Micro-segmentation enhances security.
- Network functions are decoupled from hardware.
- Enables rapid changes and policy automation.

```
4. Overall Agility Benefits (SDDC vs Traditional DC)
```

Aspect	Traditional Datacenter	Software Defined Datacenter
Provisioning Time	Hours/Days	Minutes/Seconds
Resource Utilization	Low to Moderate	High (Resource Pooling)
Scalability	Limited	Highly Scalable
Automation	Manual/Complex	Policy-driven Automation
Flexibility	Hardware Dependent	Software Defined
Agility	Low	High

```
Conclusion: SDDC enables rapid provisioning, elastic scaling, and efficient resource management, significantly improving agility compared to traditional datacenters.
```

```
sddcuser@sddc-demo:~$
```

PROBLEM 5. Multi-Cloud Federation: Identity and Privacy Design

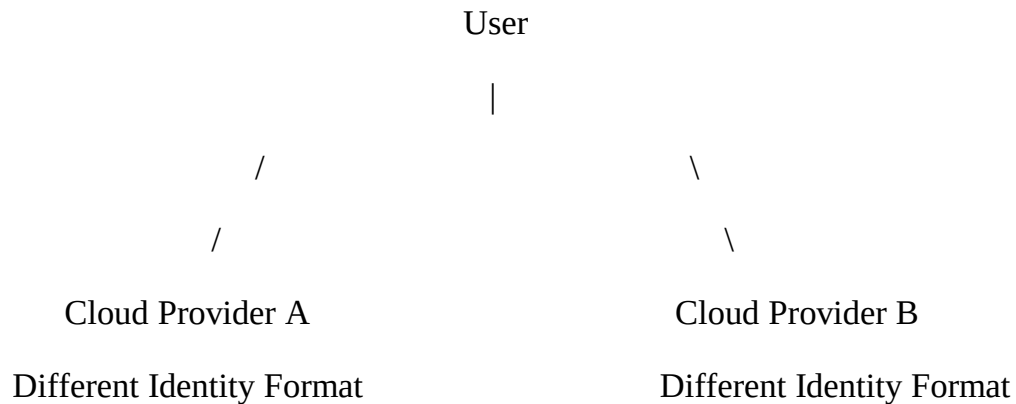
Problem Analysis

A multi-cloud federation uses multiple cloud providers. The project failed because different providers may use different:

- Identity formats

- Authentication mechanisms
- User attributes
- Access-control models
- Security policies

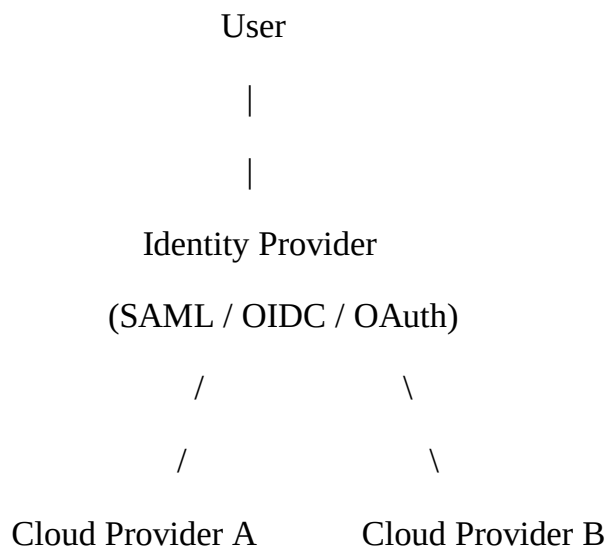
For example:



This creates **identity mismatches**.

Proposed Secure Federation Design

Use a trusted **Identity Provider (IdP)** between users and cloud providers.



Key Design Principles

1. Use Federation Standards

Use common standards such as:

- **SAML 2.0**
- **OAuth 2.0**
- **OpenID Connect (OIDC)**

These allow identity information to be exchanged securely between different providers.

2. Single Sign-On (SSO)

The user authenticates once with the trusted Identity Provider.

The user can then access authorized cloud services without repeatedly providing credentials.

3. Attribute Mapping

Different providers may use different attribute names.

For example:

Provider A: employee_id

Provider B: user_id

An identity federation layer can map these attributes into a common identity model.

4. Privacy Protection

Only the minimum required user information should be shared.

For example, if a cloud service only needs to know that a user is an employee, it should not receive unnecessary personal information.

5. Strong Authentication

Use:

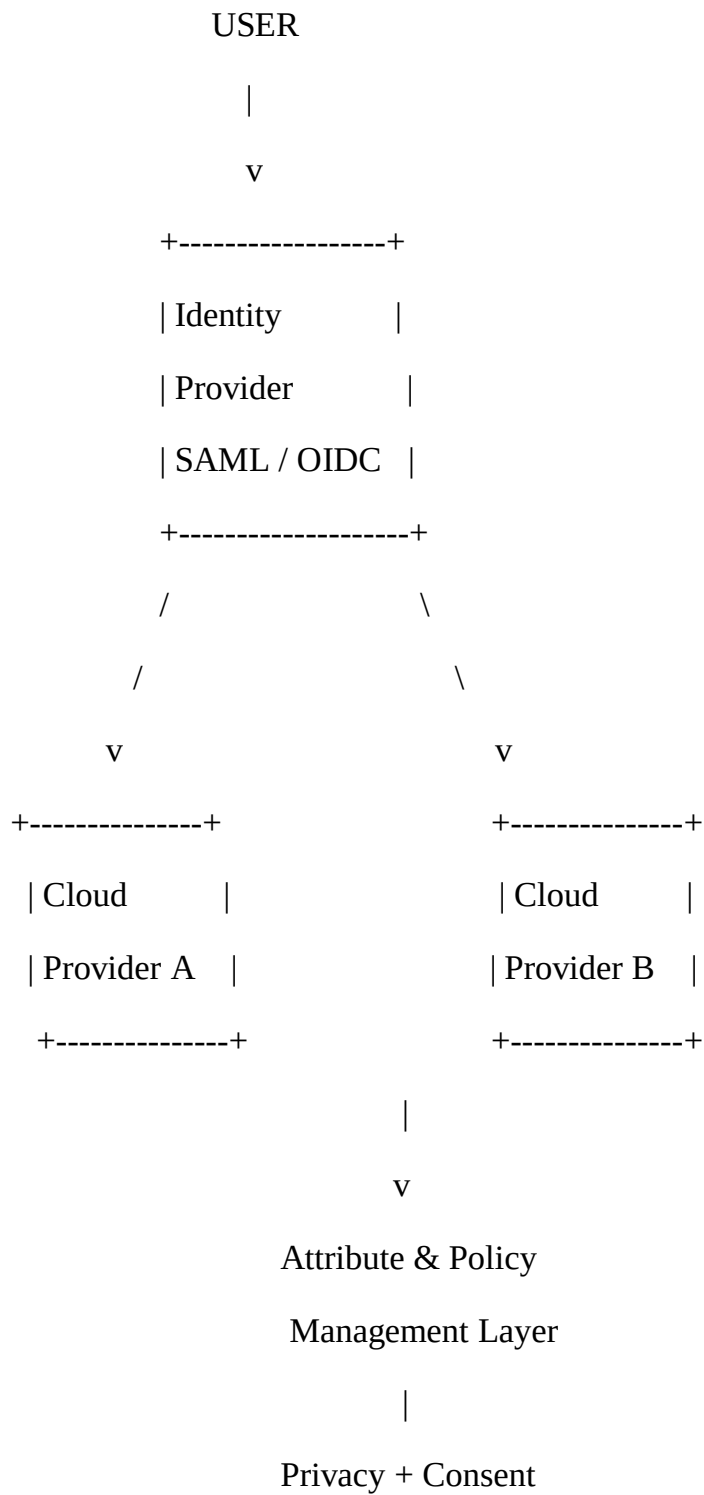
- Multi-factor authentication
- Strong authentication policies
- Short-lived tokens
- Secure session management

6. Trust Management

Establish trust between the Identity Provider and cloud providers using:

- Certificates
- Secure tokens
- Key management
- Defined federation policies

Proposed Architecture:



Conclusion

The identity mismatch can be solved through a **federated identity architecture** using a trusted Identity Provider, SAML/OIDC, SSO, attribute mapping, strong authentication and privacy-aware data sharing.

This provides **secure interoperability, centralized identity management and improved privacy across multiple cloud providers.**

```
federation@federation-demo:~$ echo "=== MULTI-CLOUD FEDERATION IDENTITY ANALYSIS & SECURE DESIGN ==="
=== MULTI-CLOUD FEDERATION IDENTITY ANALYSIS & SECURE DESIGN ===

1. Issue Analysis (Reason for Failure)
- Each cloud provider (AWS, Azure, GCP) uses different identity systems.
- User identities are not consistent across providers (e.g., alice@aws vs alice@azure).
- No trust relationship established between identity providers.
- Tokens and policies are not interoperable.
- Result: Authentication failures and authorization mismatches.

2. Impact
- Users cannot access resources seamlessly across clouds.
- Increased security risks due to weak identity mapping.
- Compliance and privacy requirements are violated.

3. Secure Identity & Privacy Design Using Federation Concepts
- Use a trusted Identity Provider (IdP) for all users.
- Implement SAML 2.0 / OpenID Connect for federation.
- Establish trust between IdP and each cloud via metadata exchange.
- Use Short-lived tokens (OAuth 2.0 / OIDC) for secure access.
- Apply Least Privilege and Role-Based Access Control (RBAC).
- Encrypt tokens and use TLS for all communications.
- Log and audit all identity and access events.

4. Proposed Architecture (High Level)
+-----+
| Users | <----- Trust (SAML/OIDC) -----> | Identity Provider |
+-----+
|
| Finance-VM (AWS) | Healthcare-VM (Azure) | Student-VM (GCP) |
| (Service SP) | (Service SP) | (Service SP) |
+-----+

5. Privacy Considerations
- Share only necessary identity attributes (minimal claims).
- Use anonymization/pseudonymization for analytics.
- Enforce data residency and compliance policies.
- Provide user consent and transparency.

Conclusion: Federation with a trusted IdP ensures seamless, secure, and privacy-preserving access across multiple clouds.
federation@federation-demo:~$
```